

Why the Item Side First: Taste Is Revealed, Not Stated

SAMPLE (thesis-write Phase C, 2026-08-20) | discussion material, not thesis prose. Follows the 3-paragraph outline agreed 2026-08-20.

An interaction occurs when the content of an item meets the taste of a user, and the bilinear score of matrix-factorization models mirrors this shape: a taste vector multiplied by a content vector [1]. The two ingredients differ in how well they are observed. The content of an item is recorded; the catalogue states its department, colour, and pattern. The taste of a user is latent. Users do not state their taste, and in most cases could not articulate it if asked. The stated attributes that are available, such as age or newsletter subscription, are a loose proxy: sufficient signal for popularity-based fallbacks, but far too unspecific to ground an explanation in. A recommendation justified with “because you are 34 and receive the newsletter” attributes the purchase to facts that did not cause it.

Taste is nevertheless not out of reach. It cannot be collected as a stated feature, but it is revealed through behavior [2]: the purchase history records what a user’s taste selects. Collaborative filtering exploits exactly this, and the learned user embedding is a summary of revealed taste. The user side therefore has no capture problem. Its problem is that revealed taste arrives without labels, as a latent vector with no vocabulary attached. For grounding, the consequence is to stop searching for taste among user attributes and instead express the captured taste in the vocabulary of the items it selects.

The same observation dictates where grounding must begin. In pure collaborative filtering, both embedding spaces are behavior summaries; the embedding of an item encodes who bought it, not what it is, and whatever legibility the item side appeared to have was borrowed from the catalogue vocabulary outside the model. The item side owns a rich recorded vocabulary that can be moved inside the model, while the user side offers only the thin stated fields discussed in the Data chapter. Grounding therefore starts on the item side. The order is forced by the data, and the user side follows once its revealed taste can be expressed in item vocabulary (the fU chapter).

References

- [1] Y. Koren, R. Bell, and C. Volinsky. Matrix factorization techniques for recommender systems. *Computer*, 42(8):30–37, 2009.
- [2] P. A. Samuelson. Consumption theory in terms of revealed preference. *Economica*, 15(60):243–253, 1948.